Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Forces & Newton's Laws

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. A particle of mass 2 kg is acted on by a force of 10 N. Find its acceleration. **[2]**

2. A particle of mass 5 kg has acceleration $3\,\text{m}\,\text{s}^{-2}$. Find the resultant force. **[2]**

3. A force of 12 N acts on a particle of mass 4 kg. Find the acceleration. **[2]**

4. A particle of mass 3 kg is at rest on a smooth horizontal surface. A horizontal force of 6 N is applied. Find the acceleration and the distance travelled in 4 s. **[5]**

5. A particle of mass 2 kg is pulled up a smooth plane inclined at 30° to the horizontal by a force of 20 N parallel to the plane. Find the acceleration. **[5]**

6. A particle of mass 4 kg rests on a smooth plane inclined at 20°. Find the component of weight down the plane and the acceleration if released. **[4]**

7. Two forces of 5 N and 12 N act at right angles. Find the magnitude of the resultant. **[3]**

8. A particle of mass 5 kg is acted on by forces $3\mathbf{i} + 4\mathbf{j}$ and $2\mathbf{i} - 3\mathbf{j}$ (in newtons). Find the acceleration. **[4]**

9. A particle of mass 2 kg is at rest. It is acted on by a constant force of 8 N for 3 s. Find the final speed and distance travelled. **[4]**

10. A lift of mass 800 kg accelerates upwards at $1.5\,\text{m s}^{-2}$. Find the tension in the cable ($g = 9.8$). **[4]**

11. A lift of mass 600 kg descends with acceleration $0.5\,\text{m s}^{-2}$. Find the tension in the cable. **[4]**

12. A particle of mass 3 kg is held at rest on a smooth plane inclined at $40°$. Find the magnitude of the force parallel to the plane needed to keep it at rest. **[4]**

13. Resolve a force of 20 N at $30°$ to the horizontal into horizontal and vertical components. **[3]**

14. A particle of mass 1 kg is subject to forces $5\mathbf{i}$ and $12\mathbf{j}$. Find the magnitude of the acceleration. **[3]**

15. A particle rests in equilibrium under three forces. Two are $3\mathbf{i} + \mathbf{j}$ and $-\mathbf{i} + 4\mathbf{j}$. Find the third force. **[3]**

Section B – Hard (Q16–25)

16. A particle of mass 5 kg is pulled up a smooth plane inclined at $30°$ by a force of 40 N parallel to the plane. Find the acceleration and the distance travelled from rest in 3 s. **[6]**

17. A particle of mass 2 kg is projected up a smooth plane inclined at $25°$ with speed $8\,\text{m s}^{-1}$. Find how far up the plane it travels before coming to rest. **[6]**

18. A body of mass 10 kg is on a rough horizontal surface. A horizontal force of 30 N produces acceleration $2\,\text{m s}^{-2}$. Find the frictional force. **[4]**

19. A particle of mass 4 kg moves under the action of forces $6\mathbf{i} - 2\mathbf{j}$ and $-\mathbf{i} + 5\mathbf{j}$. Find the acceleration and the magnitude of the resultant force. **[5]**

20. A lift of mass 500 kg contains a person of mass 70 kg. The lift accelerates upwards at $2\,\text{m s}^{-2}$. Find the tension in the cable and the reaction of the floor on the person. **[6]**

21. A particle of mass 3 kg is on a smooth plane inclined at 35°. A force of 25 N acts up the plane. Find the acceleration. **[5]**

22. Two forces act on a particle: $F_1 = 3\mathbf{i} + 4\mathbf{j}$ and $F_2 = 5\mathbf{i} - 2\mathbf{j}$. The particle has mass 2 kg and is initially at rest at the origin. Find its position after 2 s. **[6]**

23. A particle is in equilibrium under forces $2\mathbf{i} + 3\mathbf{j}$, $-\mathbf{i} + \mathbf{j}$ and $\mathbf{F}$. Find $\mathbf{F}$. **[3]**

24. A car of mass 1200 kg accelerates from rest to $20\,\mathrm{m\,s^{-1}}$ in 10 s on a horizontal road. Find the driving force if resistance is 200 N. **[5]**

25. A particle of mass m slides down a smooth plane inclined at angle α. Show that its acceleration is $g\sin\alpha$. **[3]**

End of Worksheet • *Show all working clearly. Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless otherwise stated.*